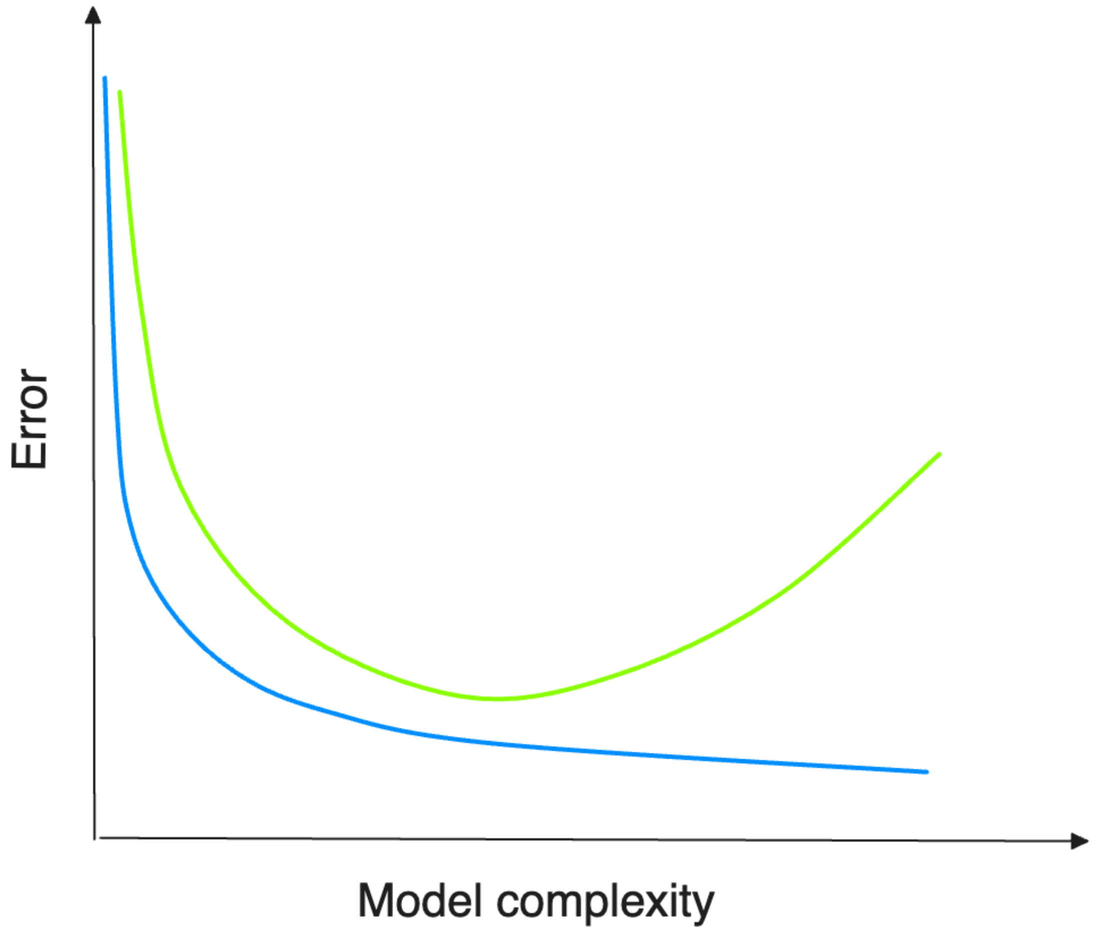
Bias Variance Tradeoff



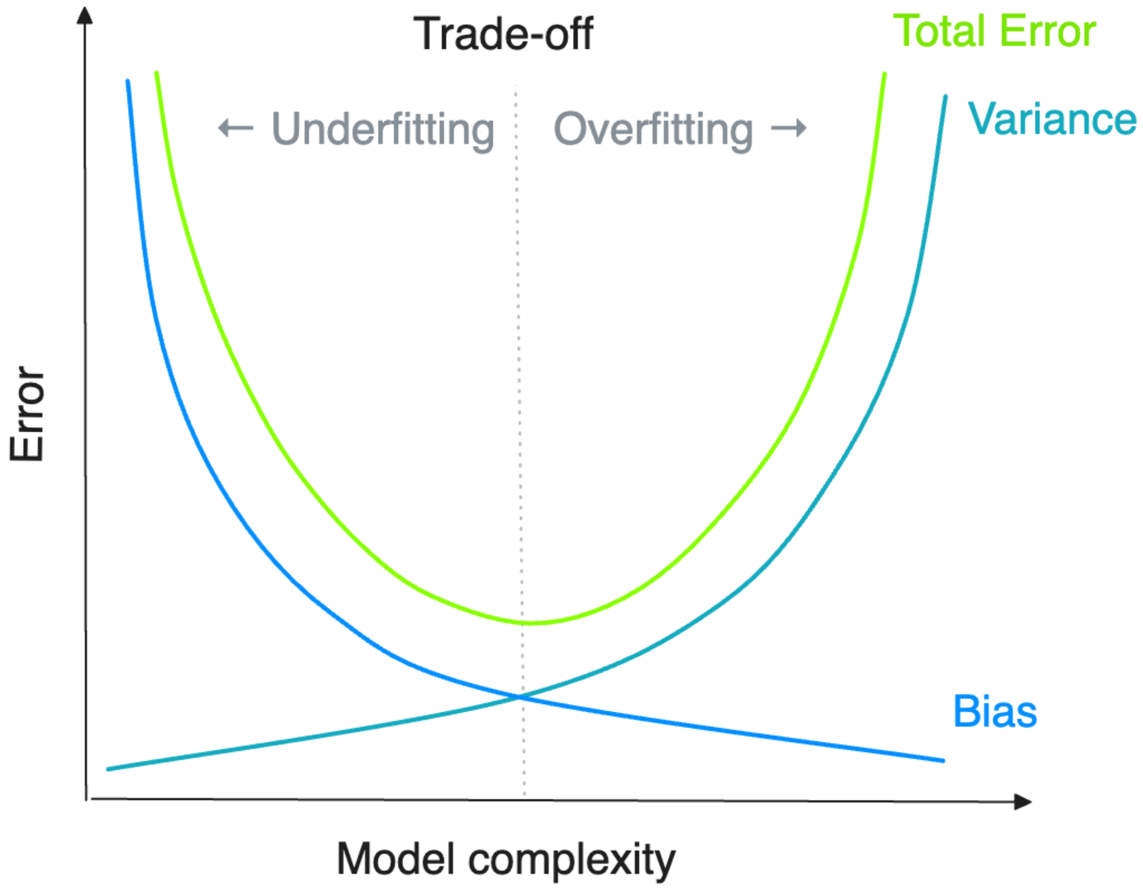
Bias and Variance

Prediction errors can be decomposed into two main components: error from bias, and error from variance[1]

Bias occurs with an underfit model, which is too simple to accurately model the relationship between input features X and some target Y

Variance arises from an overfit model that memorizes the training data points, making it unable to generalize on the unseen data

The objective is to minimize the generalization (test) error by selecting the appropriate modeling approach to find the trade-off between overfitting and underfitting[2]



1. Machine Learning University (MLU). The Bias Variance Tradeoff. https://mlu-explain.github.io/bias-variance/
2. MOOC Machine learning with scikit-learn. Bias and Variance. https://inria.github.io/scikit-learn-mooc/overfit/bias_vs_variance_slides.html

Mean Squared Error (MSE) Loss

To understand the bias and variance decomposition, we first need to look at the mean squared error (MSE), which is a loss function for regression problems (MSE also equals Brier score for binary problems)

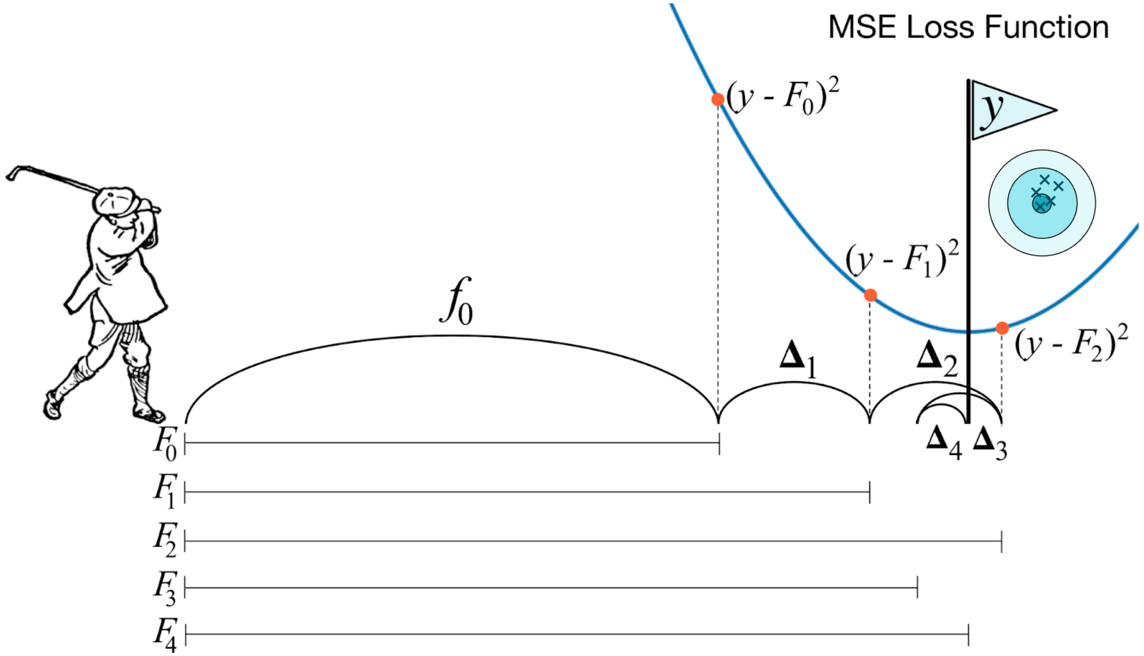
$$\mathrm{MSE} = \mathbb{E}[(y - \hat{y})^2]$$

Below is the canonical decomposition of the squared error loss into bias and variance (and some irreducible ϵ)[1]

$$\mathbb{E}[(y - \hat{y})^2] = (y - \mathbb{E}[\hat{y}])^2 + \mathbb{E}[(\mathbb{E}[\hat{y}] - \hat{y})^2]$$

$$\mathbb{E}[(y - \hat{y})^2] = [\mathrm{Bias}]^2 + \mathrm{Variance}$$

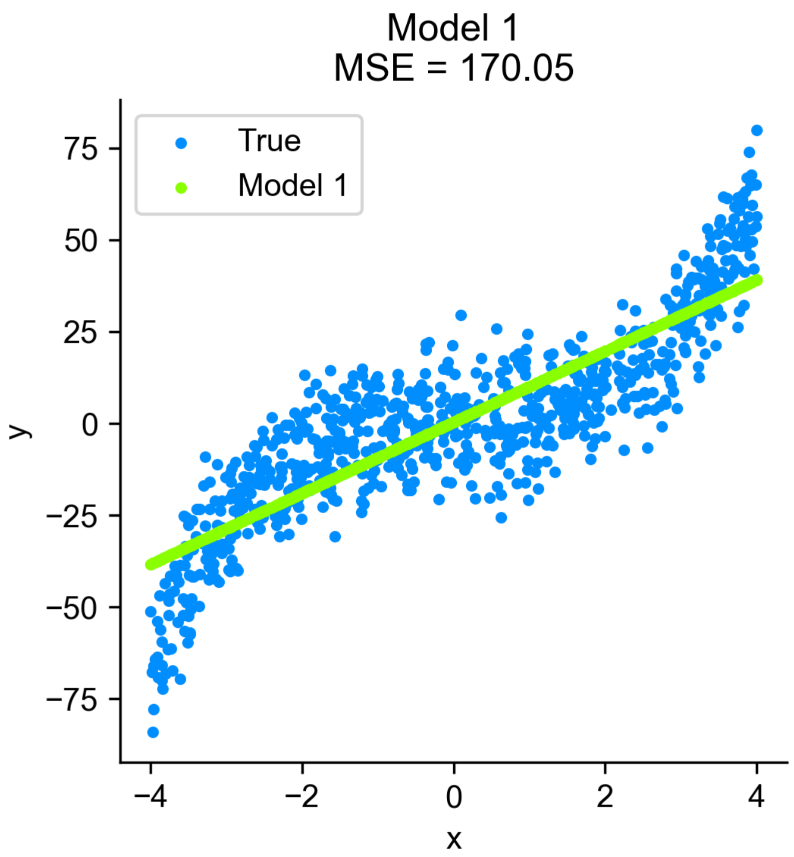
if $\mathrm{Bias}(\widehat{\theta}) = 0$ (that is, $\widehat{\theta}$ is unbiased) then $\mathrm{MSE}(\widehat{\theta}) = \mathrm{Var}(\widehat{\theta})$[2]



1. Raschka S. mxltend. bias_variance_decomp: Bias-variance Decomposition for Classification and Regression Losses. Link
2. Meyer D. A Few Notes on Fisher Information. https://davidmeyer.github.io/ml/fisher.pdf

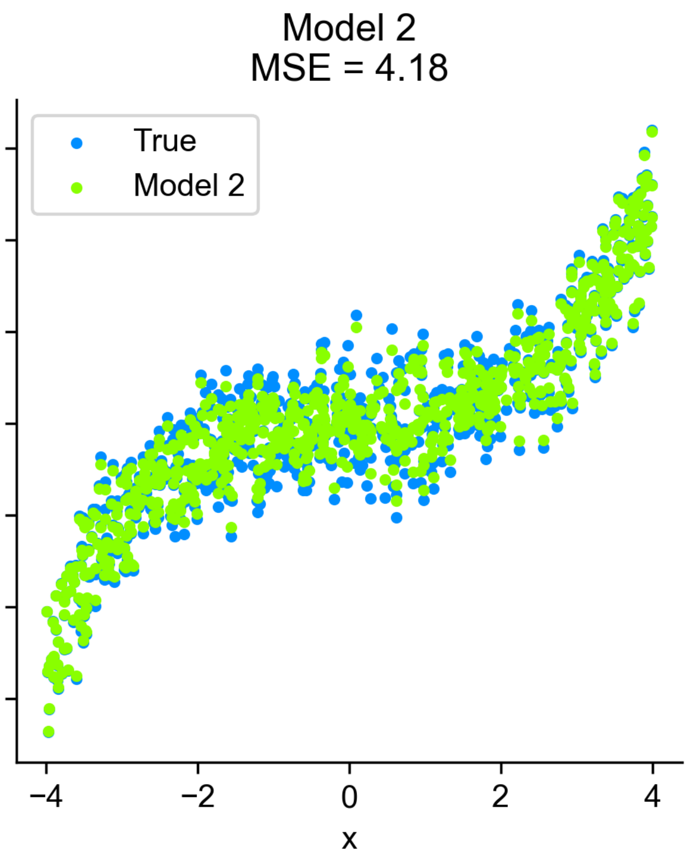
Model fit: training points

Underfitting



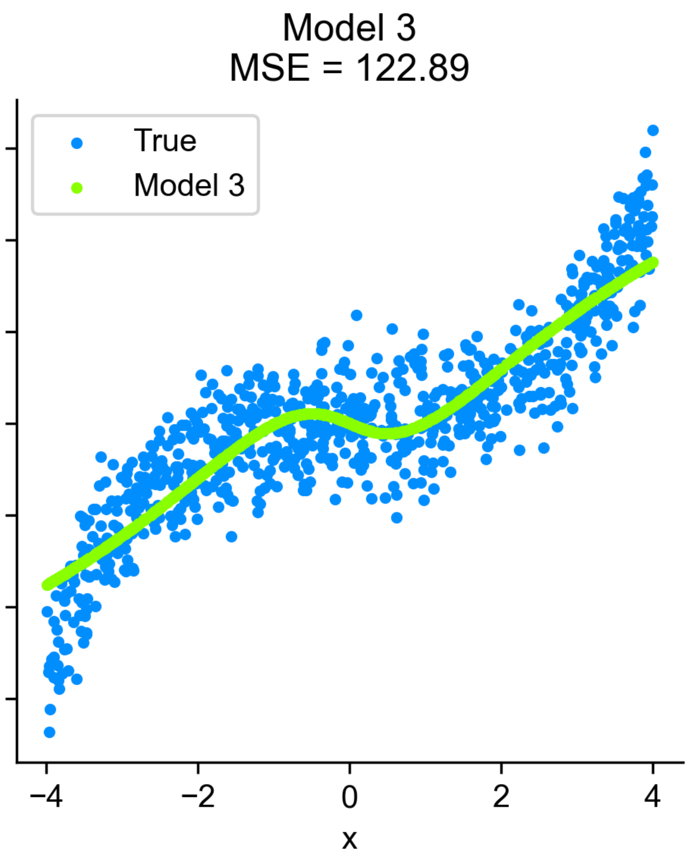
This model has too simplistic assumptions about the data

Overfitting



The model effectively memorizes the training data points

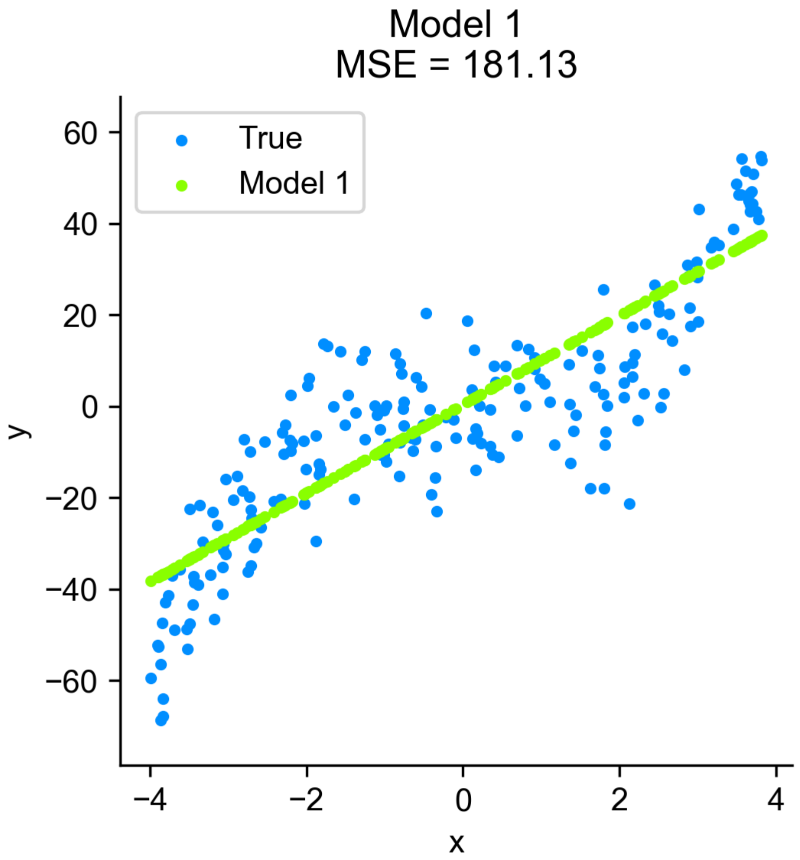
Might be just right



The model finds a general pattern of the data generating process

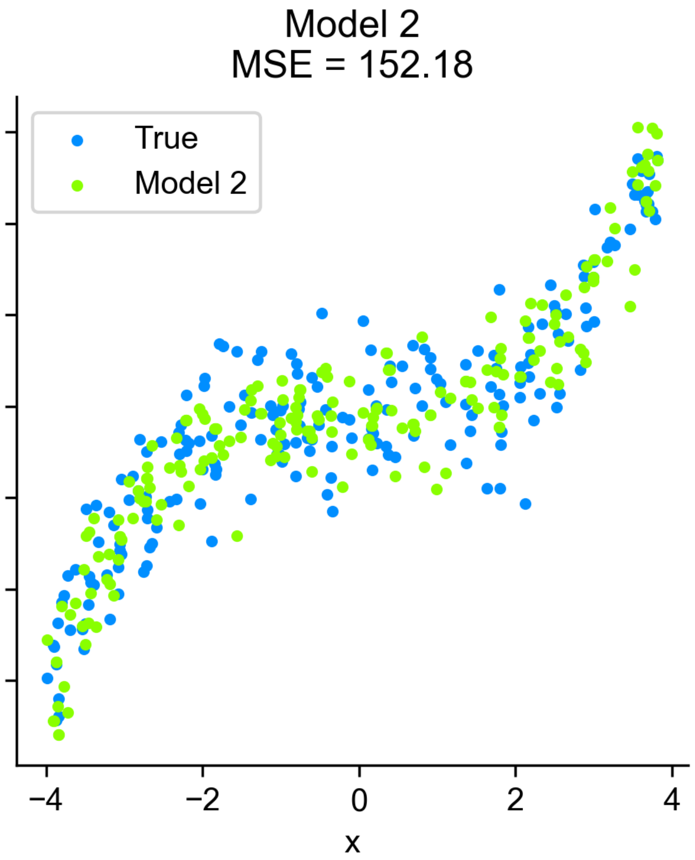
Model fit: test points

Underfitting



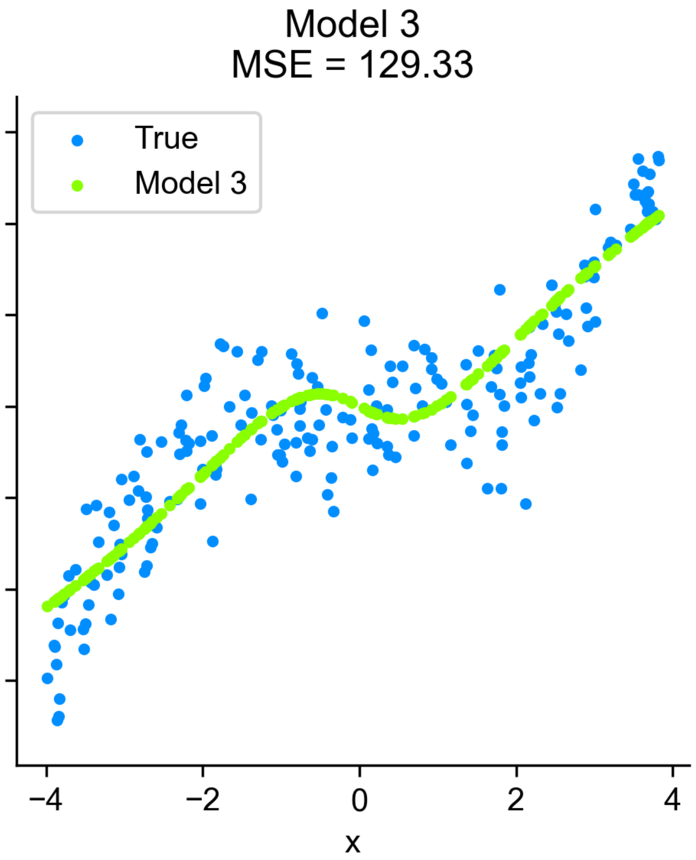
The model's error remains high for both train and test sets

Overfitting



The model has a much higher loss on the test set vs training

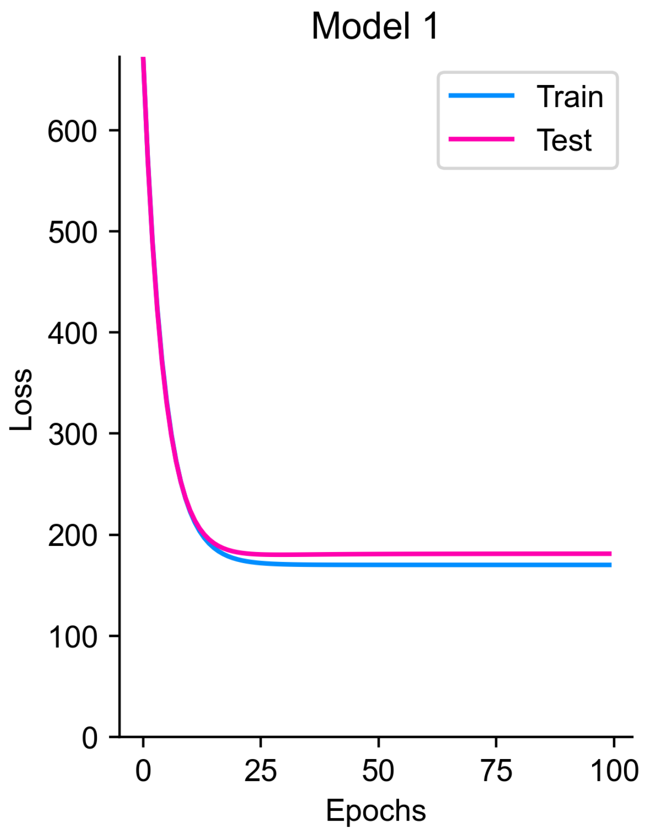
Might be just right



The general pattern remains true on the unseen data

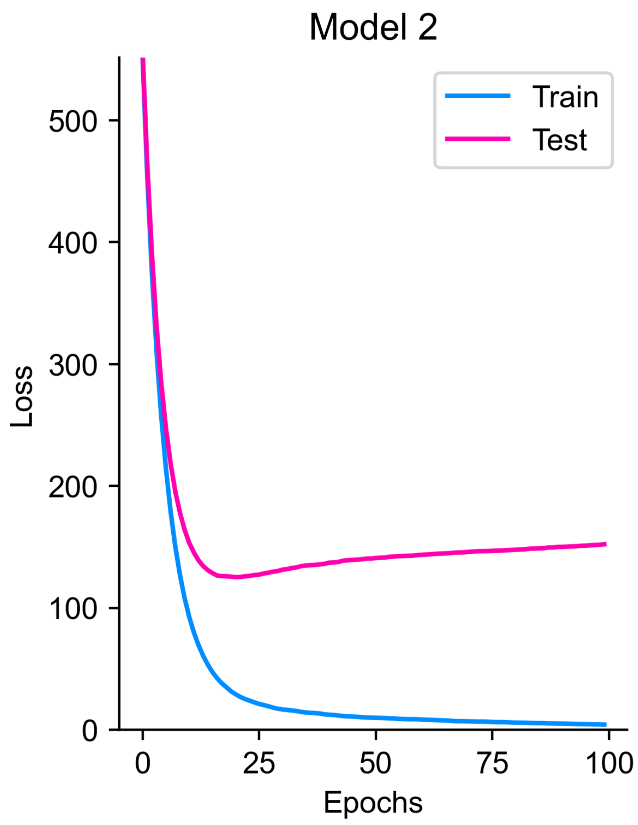
Train and test loss

Underfitting



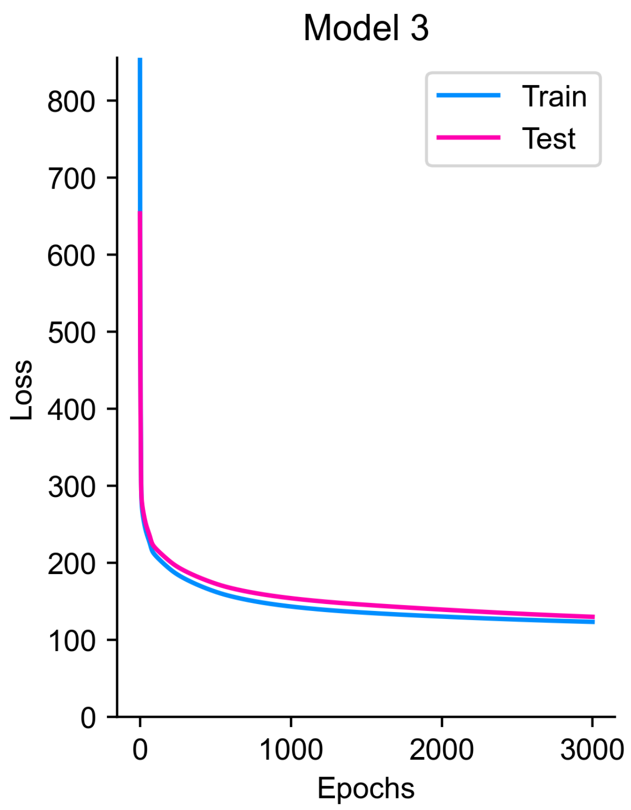
Both train and test errors are quite high

Overfitting



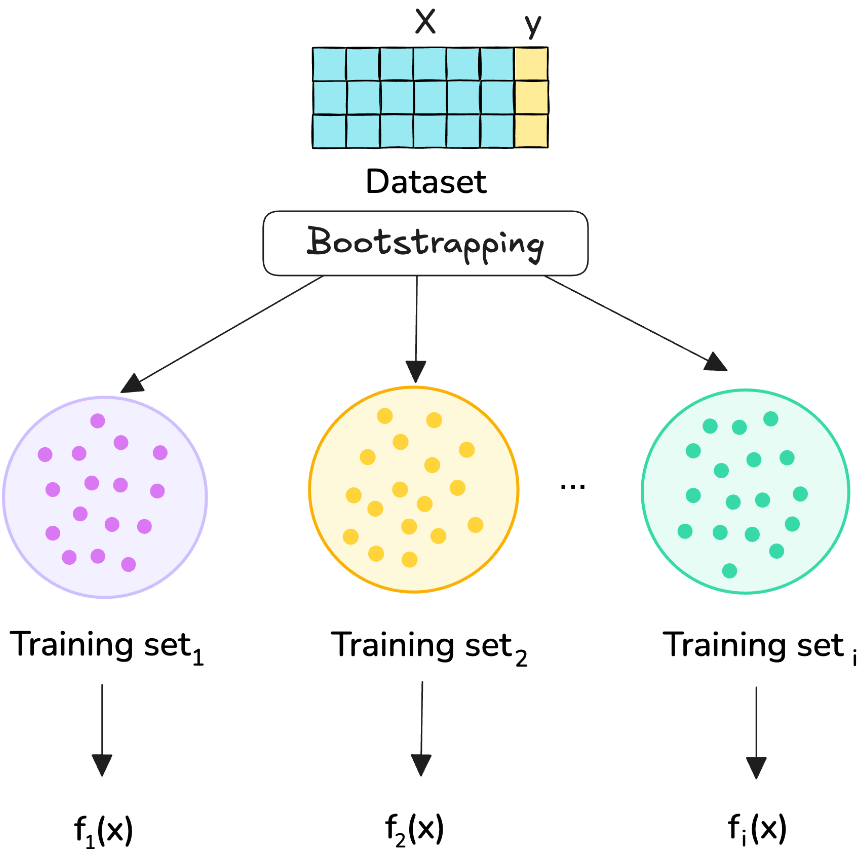
In this scenario, train loss is very low while the test loss is high

Might be just right



The model strikes a balance between underfitting and overfitting

Bias and variance decomposition



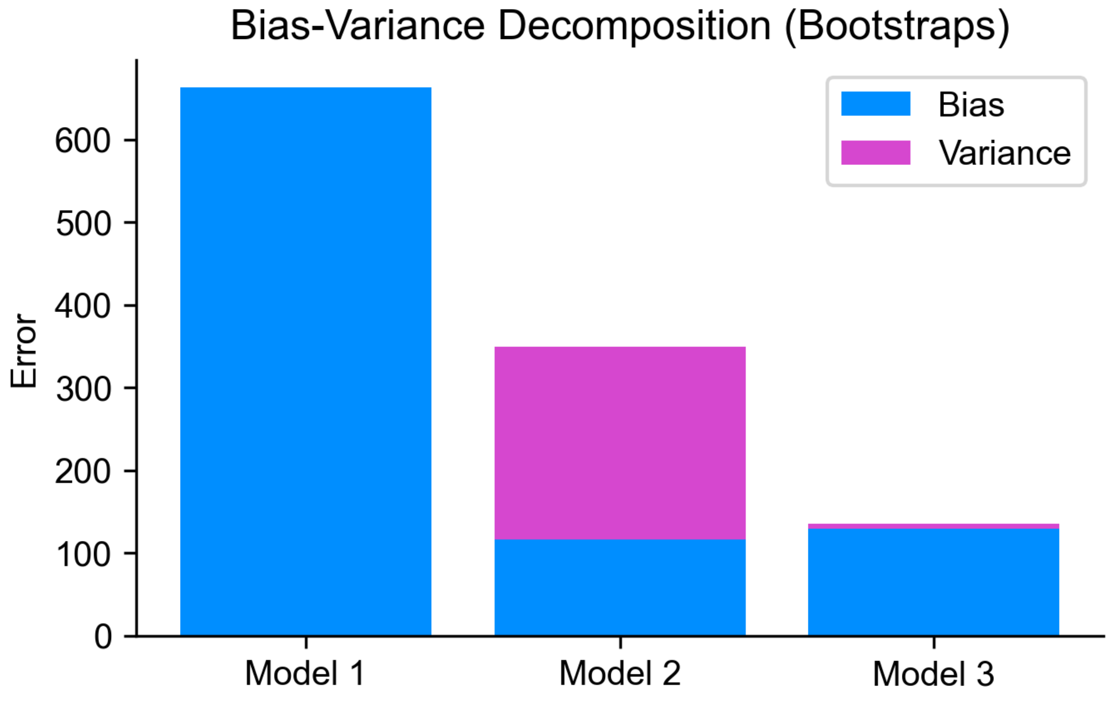
Bias = $E(y - E(f_i(x))^2$

Expected error between the target and an average of estimates

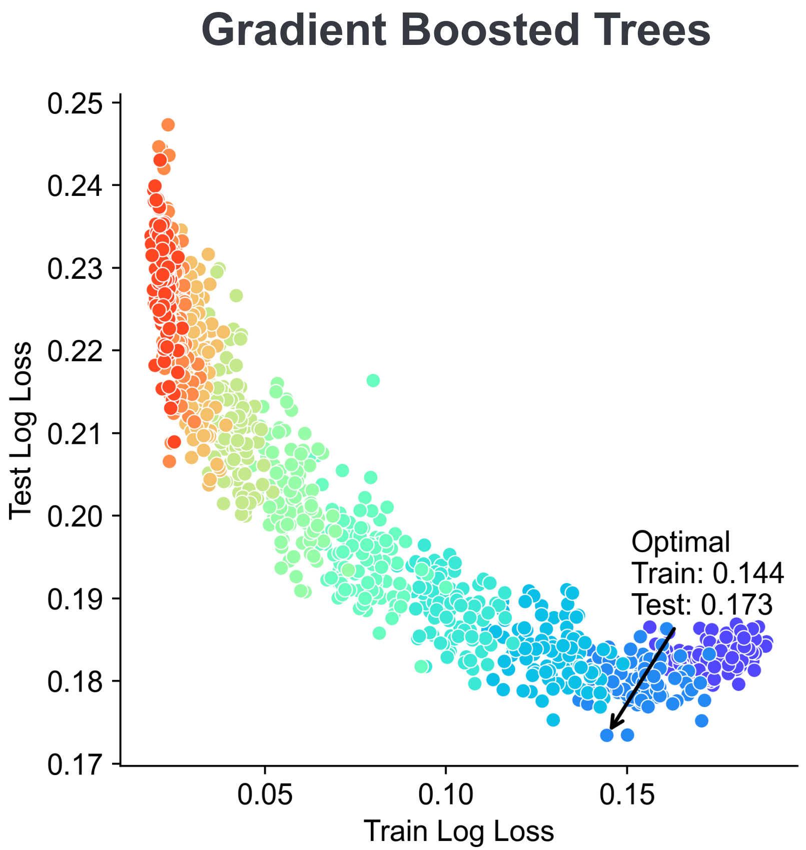
Variance = $E(f_i(x) - E(f_i(x))^2$

Average spread between the estimates compared to average estimate of models

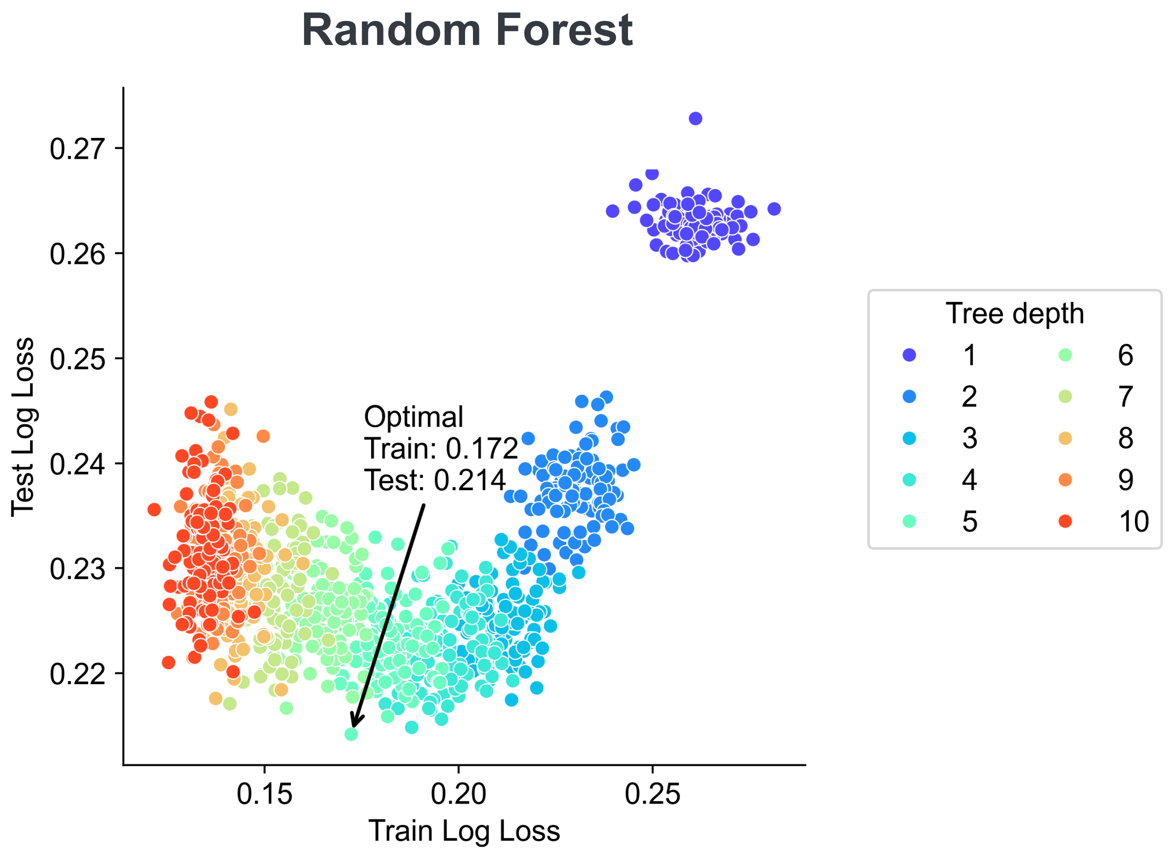
- Techniques like bootstrapping are commonly used for decomposing bias and variance by fitting models on subsets on data to derive bias and variance
- We can see that the overfitting model has the highest variance



Bias and variance for classification problems[1]



With Boosted Trees, a shallow depth around 1 or 2 is optimal for this dataset



With Random Forest, we need a larger depth to find a good balance

1. For classification problems, there's no universally accepted method for decomposing bias and variance. Here, we use bootstrapping to visualize train and test losses, helping to identify the optimal individual tree depth for tree-based models. [Link]

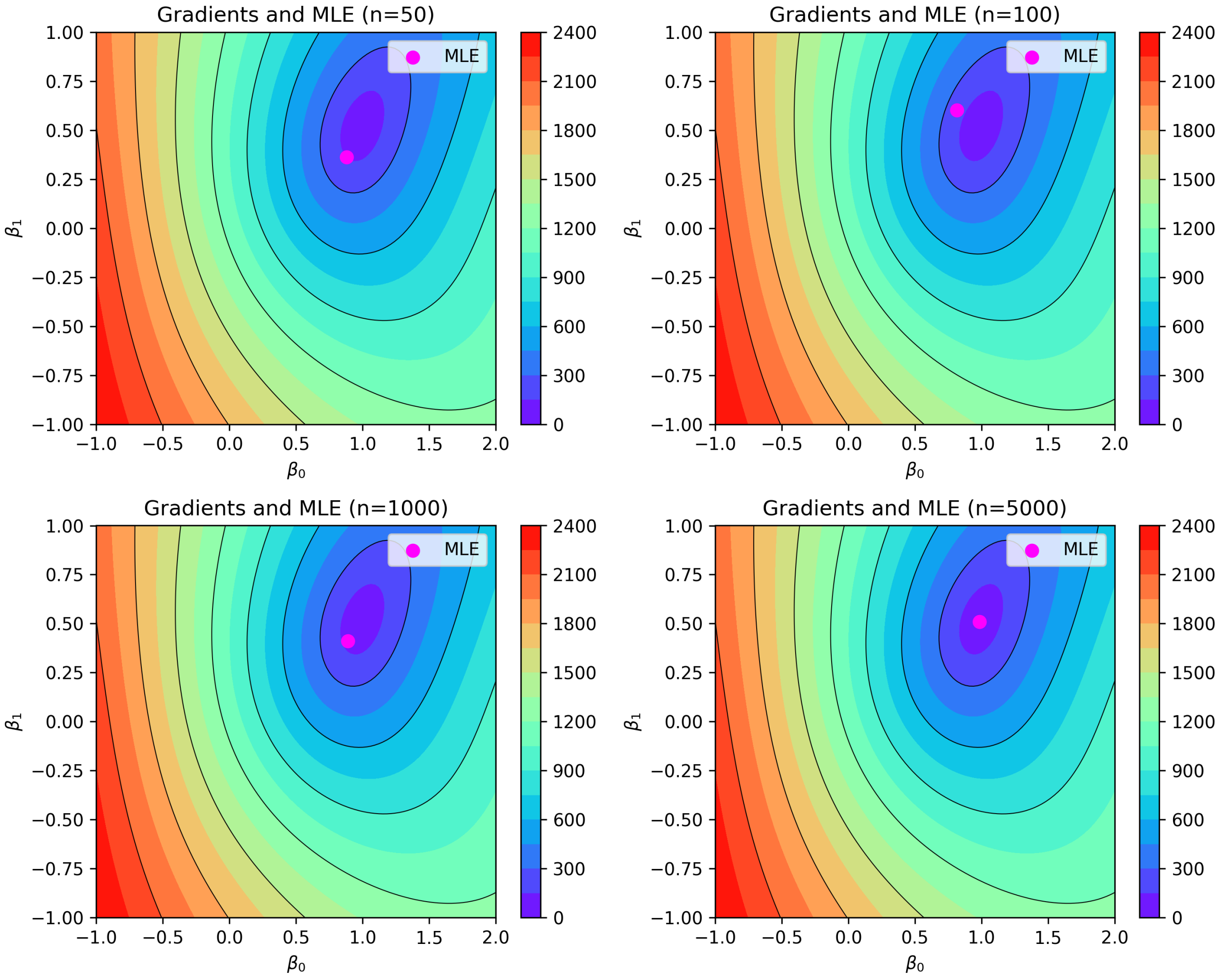
Unbiased estimators

In statistics, bias is understood as accuracy and variance as precision of an estimate, and no trade-off can be made for unbiased estimators[1]

In particular, bias is the distance between the average of the collection of estimates and the true average value of the parameter being estimated

For an unbiased estimator, the bias is equal to 0, hence the error is typically composed only of the variance component[2]

To the right, we show how with an increasing sample size, the score function (gradient) approaches the true Maximum Likelihood Estimate (MLE)



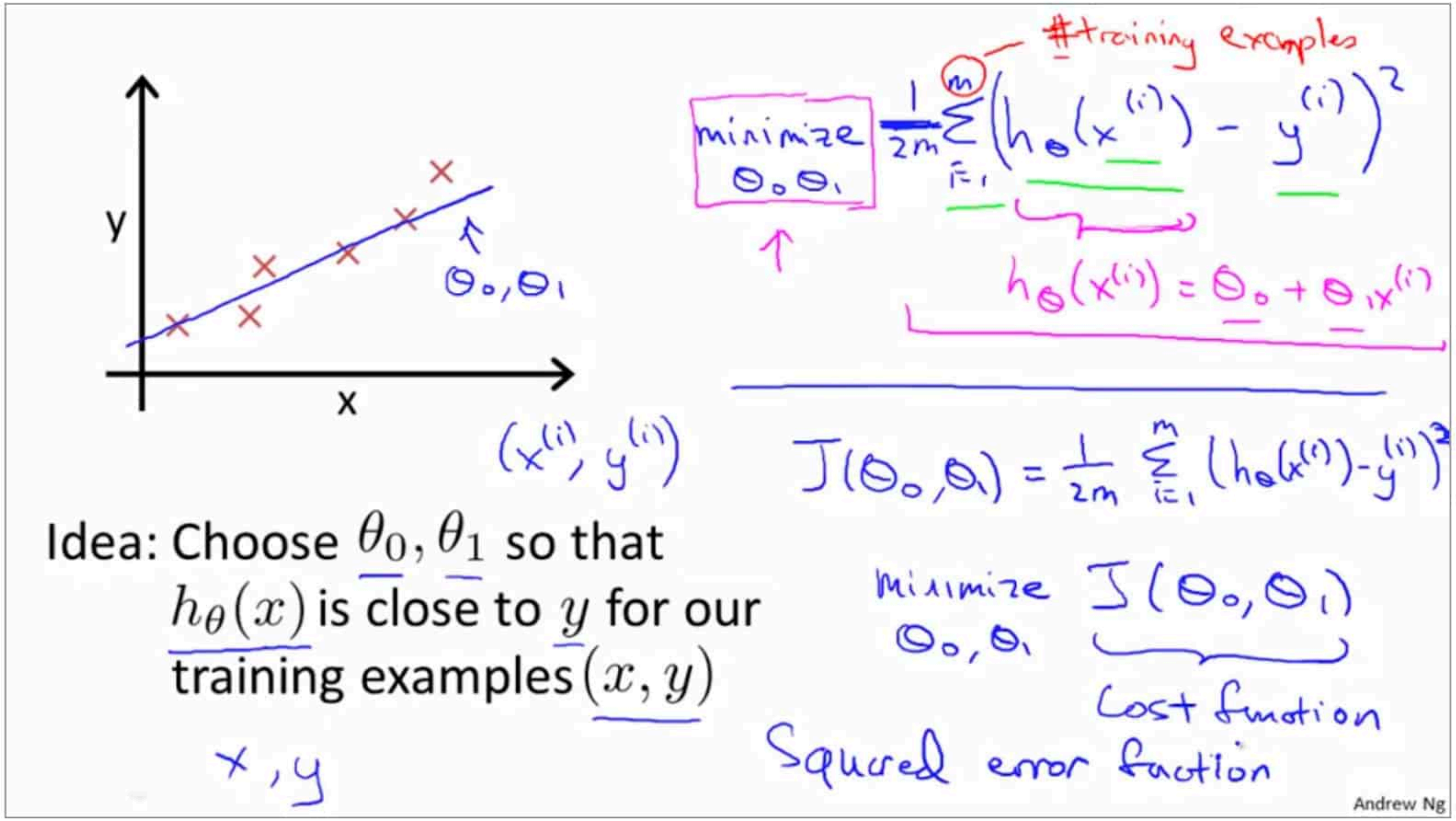
1. Lauritzen S. Properties of Estimators. https://www.stats.ox.ac.uk/~steffen/teaching/bs2siMT04/si2c.pdf
2. Meyer D. A Few Notes on Fisher Information. https://davidmeyer.github.io/ml/fisher.pdf

Notebook

Notebook with the code available here

Extras (not mine)

Cost Function

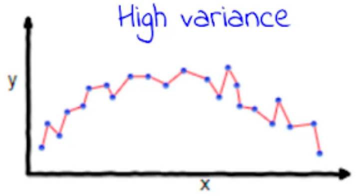


Extras (not mine)

Bias–Variance Tradeoff

Unable to generalize
Picks up dataset noise

High variance



Overfitting

Symptoms:

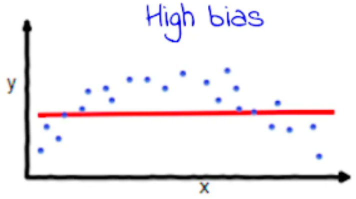
- Training error much lower than test error
- Training error lower than the expected error

Remedies:

- Add more training data
- Use simpler models
- Use bagging

Unable to capture the patterns in the data

High bias



Underfitting

Symptoms:

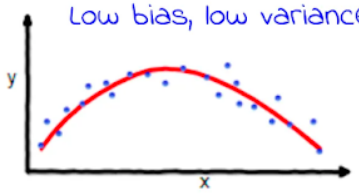
- Training error is higher than expected (often testing error too)

Remedies:

- Use more complex model (e.g. linear to non-linear)
- Use more features
- Use boosting

Captures the patterns and generalizes to test data

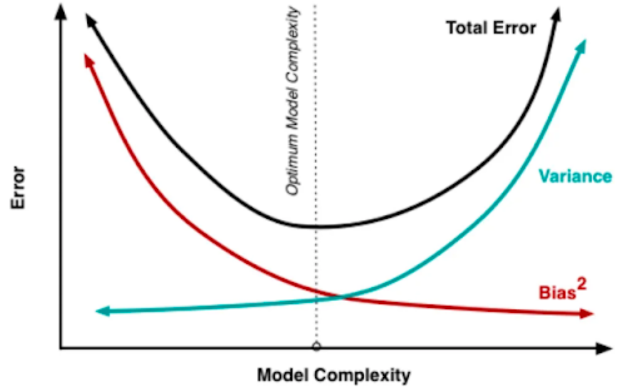
Low bias, low variance



Optimal fitting

Properties:

- Training and testing errors are similar and closer to the expected error



Tot. Err = Bias^2 + Variance + Irreducible Error

- When you try to reduce the bias too much, it increases the variance
- When you try to reduce the variance too much,, it increases the bias
- Need to find the right balance to reduce the overall error
- There will always be some error due to the noise in the dataset

References: http://scott.fortmann-roe.com/docs/BiasVariance.html | https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote12.html

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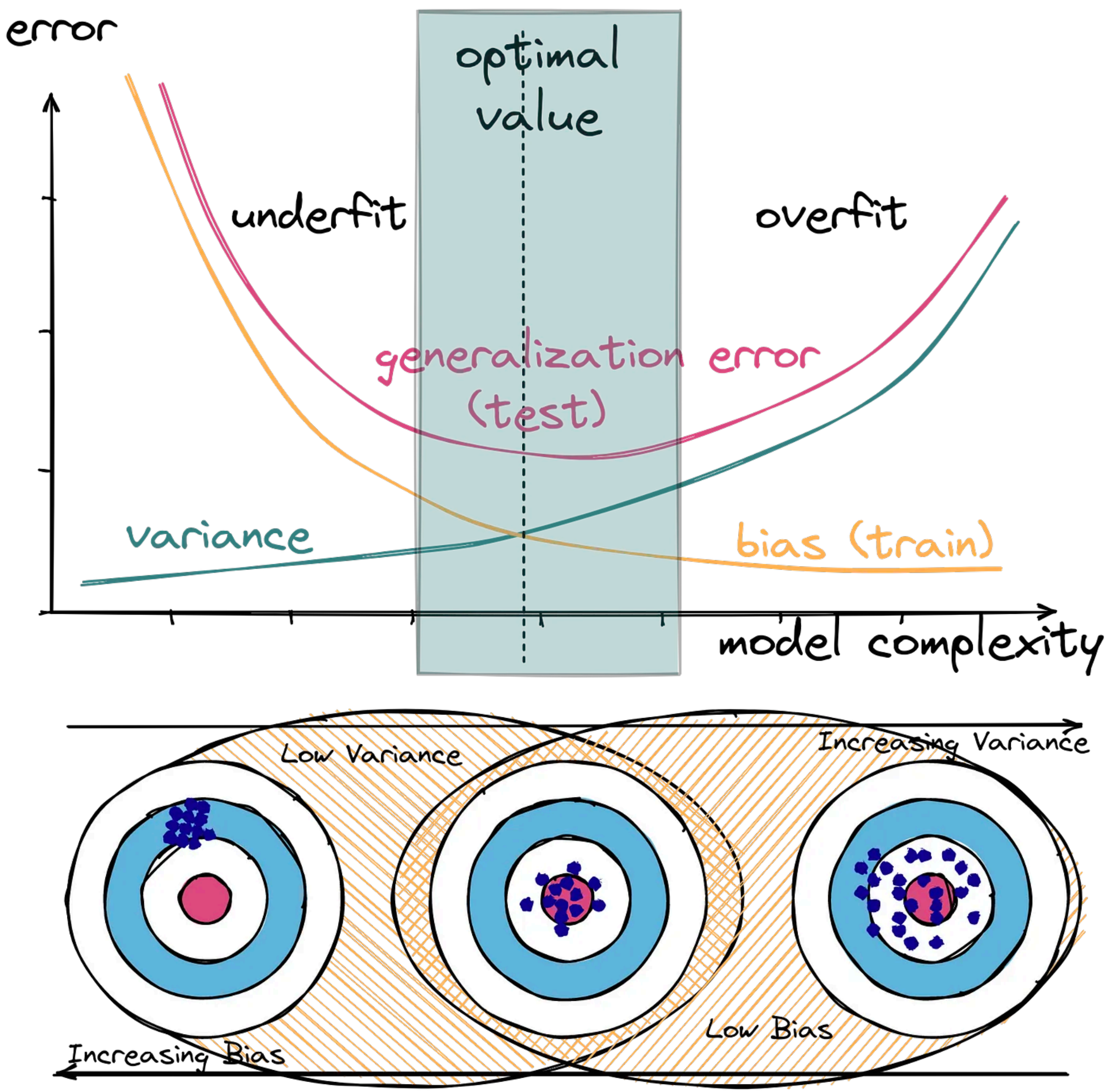

Extras (not mine)